TaniWise

AI-Powered Crop Management for
Malaysian Smallholder Farmers

BorneoHackWknd Project Report

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1. Problem Statement

Malaysia's agricultural sector is the backbone of rural livelihoods, yet smallholder farmers, particularly in states like Kedah, Perak, and Kelantan continue to face severe challenges that threaten their food security, income stability, and long-term sustainability.

1.1 The Rural Farmer’s Reality

Smallholder rice and vegetable farmers in rural Malaysia operate with limited access to modern agricultural knowledge. When crops show signs of disease, fungal infection, or nutrient deficiency, most farmers rely on trial-and-error, word-of-mouth advice, or expensive visits from agricultural extension officers. The type of responses that are often too slow and too costly to prevent significant yield losses.

Key challenges faced by these farmers include:

- Delayed crop disease identification leading to widespread crop failure and income loss.
- No early warning system for adverse weather events such as floods, droughts, or heavy rainfall that destroy harvests.
- Over-reliance on pesticides and fertilisers due to inaccurate self-diagnosis, increasing costs and environmental harm.
- Language and literacy barriers that make existing digital agricultural tools inaccessible to rural communities.
- Geographic isolation from agricultural support services, with limited road and internet infrastructure.

1.2 Scale of the Problem

According to the Department of Agriculture Malaysia, there are over 250,000 smallholder rice farmers nationwide, with a significant concentration in the MADA (Muda Agricultural Development Authority) region of Kedah — one of Malaysia's most critical rice-producing zones. These farmers contribute substantially to national food supply, yet they remain among the most economically vulnerable populations in the country.

The consequences of unaddressed crop problems are severe:

- Average crop losses of 20–40% per season due to undiagnosed diseases and pest infestations.
- Income volatility that pushes farming families into debt cycles.
- Younger generations abandoning farming, threatening the future of Malaysia's food sovereignty.
- Increased deforestation pressure as farmers expand acreage to compensate for low yields from existing land.

1.3 The Information Gap

The core of the problem is an information gap. Professional agronomy advice is expensive, infrequent, and linguistically inaccessible for many farmers who speak only Bahasa Malaysia or regional dialects. Existing digital solutions are often designed for large commercial farms, require high-speed internet, and use technical language that alienates smallholder farmers.

TaniWise was developed to bridge this gap: to put expert-level agricultural knowledge directly into the hands of every farmer, in their own language, on the device they already own.

2. Objectives

TaniWise is designed to address the challenges outlined above through a set of clearly defined objectives, aligned with the United Nations Sustainable Development Goals (SDGs).

2.1 Primary SDG Alignment: SDG 2 — Zero Hunger

TaniWise is aligned with SDG 2: Zero Hunger, and specifically targets SDG Target 2.3:

> *By 2030, double the agricultural productivity and incomes of small-scale food producers, in particular women, indigenous peoples, family farmers, pastoralists and fishers, including through secure and equal access to land, other productive resources and inputs, knowledge, financial services, markets, and opportunities for value addition and non-farm employment.*

TaniWise directly operationalises this target by democratising access to agricultural knowledge, increasing productivity through timely disease detection, and creating a pathway to financial services for smallholder farmers.

2.2 Project Objectives

Objective 1: Enable Early and Accurate Crop Disease Detection

Develop an AI-powered image recognition system that allows farmers to photograph their crops and receive an instant, accurate diagnosis of diseases, pest infestations, or nutrient deficiencies without requiring specialist knowledge or an agronomist visit.

Objective 2: Provide Accessible Agricultural Guidance

Deploy a bilingual (Bahasa Malaysia and English) AI chatbot that communicates agricultural advice in farmer-friendly, jargon-free language, making expert knowledge accessible to farmers of all literacy levels across Kedah, Perak, and Kelantan.

Objective 3: Deliver Localised Weather Forecasting and Alerts

Integrate localised weather forecast data to provide farmers with advance warnings of adverse conditions, enabling them to take timely protective action for their crops and plan planting and harvesting activities optimally.

Objective 4: Optimise Planting and Harvesting Timing

Use AI-driven recommendations to advise farmers on the best times to plant and harvest based on current soil conditions, weather patterns, and seasonal data — maximising yield and reducing the risk of loss.

Objective 5: Build a Scalable Platform for Long-Term Growth

Create a mobile application architecture that can scale from its Phase 1 pilot of 500 MADA rice farmers in Kedah to Phase 2 rubber and palm oil smallholders, Phase 3 market and financial integration, and ultimately an ASEAN-wide expansion.

3. Solution

TaniWise is a React Native mobile app leveraging Google Gemini AI to provide Malaysian farmers with instant crop disease diagnosis and treatment recommendations. This MVP, built for a 500-farmer MADA pilot in Kedah, uses a Node.js backend and Gemini Vision API to transform field photos into structured, actionable data for real-world deployment.

3.1 Application Overview

TaniWise translates the name into its purpose: it is an intelligent farming companion. The app is available in both Bahasa Malaysia and English, uses conversational and jargon-free language, and is designed to run efficiently on low-end Android smartphones with limited data connectivity; the devices most commonly owned by rural farmers.

The application consists of four core modules:

Module 1: AI Crop Diagnosis (Crop Scanner)

Farmers point their smartphone camera at a diseased leaf, stem, or fruit. The app uses Google Gemini's vision capabilities to analyse the image and return a clear, plain-language diagnosis within seconds. The response includes the likely cause, recommended treatment, and preventive measures for future seasons — all written in the way a trusted farming neighbour would speak, not a textbook.

Module 2: Smart Weather Alerts

TaniWise integrates localised weather data relevant to the farmer's specific district (e.g., Alor Setar, Kota Bharu, Ipoh surrounds). The app proactively sends notifications when significant weather events are forecast, giving farmers at least 24–48 hours to take protective action such as harvesting early, applying protective measures, or adjusting irrigation. Optimal planting and harvesting timing suggestions are updated dynamically based on current conditions.

Module 3: AI Farming Assistant (Chatbot)

Farmers can ask any farming-related question through a conversational interface powered by Gemini. The chatbot is trained to respond in Bahasa Malaysia or English, with responses adapted for farmers — using simple words, relatable examples, and actionable advice. Common questions addressed include fertiliser dosing, pest control, market preparation, and crop rotation guidance.

Module 4: Recommendation Engine

Based on the farmer's crop type, location, season, and historical patterns, the recommendation engine surfaces proactive suggestions — from the best time to apply fertilizer, to early indicators of disease risk based on current weather conditions. In Phase 3, this module will integrate FAMA market price data to advise on optimal selling timing and connect farmers with Agrobank products for insurance and microfinancing.

3.2 AI Technology: Google Gemini

TaniWise is built on Google Gemini AI as its core intelligence layer. The choice of Gemini was deliberate and strategic:

- **Multimodal capability:** Gemini's vision model enables image-based crop disease detection without requiring a separately trained specialist model.
- **Multilingual fluency:** Gemini supports high-quality Bahasa Malaysia responses natively, critical for accessibility in rural Malaysia.
- Gemini's efficiency on mobile devices supports low-data usage, crucial for rural areas with limited connectivity.

The following table summarises the AI components integrated into TaniWise:

AI Feature	Technology Used	Benefit to Farmer
Crop Disease Detection	Gemini (image recognition)	Instant diagnosis by photographing leaves or stems — no agronomist visit needed
Conversational AI Assistant	Gemini Chatbot (Bahasa Malaysia / English)	Jargon-free advice in the farmer's preferred language; accessible on low-end phones
Localized Weather Alerts	Gemini + weather API integration	Advance warnings for rain, drought, or storms to protect crops and plan activities
Planting & Harvesting Timing	Recommendation engine (Gemini)	AI-optimised timing advice based on current soil, weather, and seasonal data

3.3 Accessibility & Inclusion Design

Accessibility was a first-principles design consideration, not an afterthought. TaniWise was built with the following inclusive design principles:

- Dual-language interface: Full Bahasa Malaysia and English support throughout the application, with automatic language detection.
- Farmer-friendly language: All AI responses are prompted to avoid agricultural jargon and use analogies and language familiar to rural communities.
- Low-data mode: Core features are optimised for 3G connectivity or lower; offline caching is implemented for recently viewed disease diagnoses and weather forecasts.
- Low-end device compatibility: The application is built to perform on Android devices with 2GB RAM, covering the majority of smartphones owned by rural farmers.
- Simple UX: Large touch targets, icon-based navigation, and voice input support reduce the literacy barrier for app interaction.

3.4 Sustainability Considerations

TaniWise is designed not only to solve an immediate problem, but to create lasting, positive change across environmental, economic, and social dimensions.

Environmental Sustainability

- Reducing crop loss and food waste: By enabling early disease detection, TaniWise helps prevents crop losses of 20–40% per season, reducing the amount of food wasted before it reaches the market.
- Reducing chemical over-use: Accurate diagnosis eliminates the common practice of applying pesticides and fungicides indiscriminately, reducing environmental contamination and input costs.

- Climate-adaptive farming: Weather alerts and timing recommendations help farmers adapt their practices to changing climate patterns, building long-term resilience.
- Reduced deforestation pressure: Higher yields from existing land reduce the incentive to clear new forest areas, directly contributing to Malaysia's EUDR compliance obligations (particularly relevant in Phase 2).

Economic Sustainability

- Lower input costs: Precision recommendations mean farmers spend less on pesticides, fertilisers, and emergency interventions.
- Higher yields and income: Early disease detection and optimised timing improve harvest quality and quantity, directly increasing farmer income — contributing to SDG 2.3's goal of doubling smallholder productivity.
- Market access: Phase 3 integration with FAMA price data empowers farmers to sell at optimal times, capturing better prices and reducing exploitation by intermediaries.
- Financial inclusion: Agrobank integration creates a pathway for smallholders to access crop insurance and microloans, reducing vulnerability to single-season shocks.

Social Sustainability

- Knowledge democratisation: Expert agricultural knowledge that was previously available only to well-connected or wealthy farmers is made universally accessible.
- Community resilience: As more farmers in a region adopt TaniWise, community-level data on disease outbreaks and weather patterns can be aggregated to provide early warnings at scale.
- Youth engagement: A modern, digital tool may attract younger Malaysians back to farming careers, addressing the ageing farmer demographic that threatens long-term food production.

3.5 AI Acknowledgement

TaniWise is an AI-first product. The following acknowledges the role, limitations, and responsible use of AI in this solution:

AI Tools Used in Development

- Google Gemini API (core AI engine): Image recognition, conversational AI, recommendation engine, and multilingual response generation.

Responsible AI Considerations

- Accuracy validation: AI diagnoses are framed as recommendations, not definitive medical-grade diagnoses. The app encourages farmers to seek a second opinion from the Department of Agriculture for unusual or severe cases.
- Data privacy: Crop images and location data submitted by farmers are processed securely. No personally identifiable farmer data is stored or sold. Aggregated, anonymised data may be used to improve model accuracy over time.
- Bias mitigation: The Gemini model is prompted with Malaysian-specific crop types, diseases, and conditions to minimise the risk of advice calibrated to non-local farming contexts.
- Human oversight: TaniWise is positioned as an augmentation tool for farmers and extension officers, not a replacement for human agricultural expertise.

4. Technical Implementation

This section outlines the technical foundation of TaniWise — the tools, APIs, and architectural decisions that power the application. Every technology choice was guided by one principle: building something that works for a farmer in Kedah with a low-end Android phone and an intermittent internet connection.

4.1 Development Stack

TaniWise was built using a modern, cross-platform mobile development stack chosen for speed of development, wide community support, and suitability for rural-facing applications.

Category	Tool / Technology	Why We Chose It
Language	TypeScript & JavaScript	Type-safe codebase ensures reliability; large ecosystem of libraries for rapid MVP development
Mobile Framework	Expo (React Native)	Single codebase deploys to both Android and iOS; Expo Go enables rapid testing without full app store builds
IDE	VS Code & Android Studio	VS Code for application logic and API integrations; Android Studio for device emulation and Android-specific testing
Version Control	GitHub	Collaborative development across team members with branching strategy for feature isolation and code review

4.2 APIs & Services

TaniWise integrates four Google APIs, each chosen to address a specific farmer need. The table below maps each API to its function within the app and the direct benefit it delivers to the end user.

API / Service	What It Does in TaniWise	Farmer-Facing Benefit
Gemini API	The system processes images to generate corresponding outputs. It also features a voice-based Q&A chatbot and translates complex, technical disease diagnoses into simple, everyday Bahasa Malaysia.	Advice feels like talking to a knowledgeable neighbour; farmers can ask questions by voice in Bahasa Malaysia
Text-to-Speech API	Reads all recommendations, diagnoses, and alerts aloud in Bahasa Malaysia for farmers with reading difficulties	Elderly farmers with limited literacy can still receive and act on all app guidance — no reading required
Google Maps Platform	Field mapping, location-based soil type identification, and nearest agricultural input supplier lookup	Recommendations tailored to the farmer's specific land and location; farmers can find the nearest shop for supplies
Firebase	Offline-first data storage with automatic sync when connectivity returns; push notifications for weather alerts	App remains fully functional without internet — critical for farmers in rural areas with patchy connectivity

4.3 How It All Works Together

The following describes the end-to-end flow when a farmer uses the crop diagnosis feature — the most technically integrated part of TaniWise:

1	Farmer captures a photo — using their smartphone camera within the Expo app, the farmer photographs a leaf, stem, or fruit showing signs of disease or distress.
2	Gemini processes the technical diagnosis with a prompt to translate the findings into colloquial Bahasa Malaysia. This results in a friendly, jargon-free explanation with clear next steps for the user.
3	Text-to-Speech reads it aloud — the final response is rendered as text on screen and simultaneously read aloud, ensuring elderly or low-literacy farmers can receive the advice fully.
4	Firebase caches and syncs — the diagnosis and all recommendations are stored locally via Firebase's offline-first architecture. If connectivity was lost mid-session, data syncs automatically when restored, and weather push notifications are delivered.